

Jean-François Cloutier: "An experiment in artificial agency"

1:12:22

all

right that was a great

presentation this will be the last of

the recorded presentations for the

Symposium and last video on this part

one stream we have JF cloutier an

experiment in artificial

agency an experiment in artificial

agency I'm Jean

I'm a research fellow at the Active

infant Institute and I do research in

cognitive

robotics one of principal question I ask

myself is what does it take at a minimum

for an autonomous robot to learn to

survive in a world it knows initially

almost nothing

about other questions which follow is

are can a robot act for its own

reasons does it have agency can its

agency be

programmed and would such a robot be an

active inference

agent the one way to find out is to

actually build an artificial

agent uh and by artificial agent I mean

a stance making adaptive and

self-directed

robot so this project is a continuation

of of a personal project that I started

quite a few years ago and 3 years

ago here's where I was I had a Lego

robot uh with uh quite a few

sensors um like ultrasonic uh sensors

infrared sensor um a color sensor

luminum sensor um Motors of different types all this uh running uh from a Raspberry Pi with a board that provides an interface between the Raspberry Pi and the various Lego motors and sensors and the robot also had a 360 degree

Beacon that allowed one robot to see another through uh the other robot's 360 degree Beacon

Seeker so let's look at them in action

I named them uh Carl and Andy for I think quite obvious

reasons the Carl had gone through quite a lot of training runs so it was better at finding the food which is the piece of paper on the flo um with a a beacon infer Beacon providing the scent the attractive

scent we can see that Andy is uh having a bit of a hard time

and they see each other so that if one robot starts

acting panicky which they do if they

Collide too often the other will notice

the pattern of movement and if one robot

makes a beline then the other robot will

perceive that and say aha that's where

the food is and try to move in the same

direction but we see them

uh moving about and trying to get a food

and uh eating the

food and Pa is clearly uh more apt at

this task than

andd and what they've learned through

various

trials are the policies that

um are most successful at finding the

food so that's where I was um three years ago the way I implemented uh the cognitive capabilities quote unquote of these robots was based on a uh architectural model uh described by Marvin Minsky in his book The Society of Mind this um architecture posits that interesting behaviors emerge from many simple actors that interact in simple ways so we're looking at cognition as a collective intelligence the implementation was a hierarchy of these simple processes I call them cognition actors and they formed an abstraction hierarchy with at the very bottom um cognition actors that simply wrap sensors in uh effectors Motors and then at the very bottom we have cognition actors that care about very simple things like the location of food then at a higher level information obtained from the food location cognition actor feeds into the food approach cognition actor and this one is in tries to figure out am I getting closer to the food and so forth and so on other cognition actors uh are concerned with avoiding collisions and avoiding dark places now each cognition actor as I said is a separate process and each such process uh does very simple things it makes predictions about what it will observe next it then updates its beliefs based on uh predictions that were successful

or prediction errors received and then selects a policy randomly in the beginning but then based on on uh what was successful in the past and applies that policy acts and then decides and then predicts what and what the next incoming Sensations will be and so forth and so on essentially implementing an UDA Loop observe Orient decide and act there's one big problem though that this Society of Mind these this hard of cognition actors is predefined it's hardcoded for each robot it's the same the only thing that's learned is what policy is more effective over time based on experience so the robot acted for my reasons for the programmer's reasons not its own the robots had autonomy but no agency

in 2022 I joined the active inference Institute and continued this project under the uh symbolic cognitive robotics um

tag and I asked myself what if what if a robot started with only a rudimentary Society of Mind what if it didn't have that

pre-built Society of Mind this this hierarchy of function actors it started only with the the the most most basic ones the ones that wrap sensors and Motors then the Society of Mind uh would grow from experiencing autonomous engagement so the robot would start and with all these very basic um prior cognition actors for sensing and

affecting and then would
create new cognition actors then these
cognition actors uh would be very uh um
very simple in the sense that they would
take their Sensations from the the uh
uh the sensors themselves and would act
uh by uh engaging the motors and then uh
observe how it changes its its its
perceptions of say color or distance and
whatnot

Etc so action actors would be added to
the Society of Mind bottom up to make
sense of observations and
actions now each new cognition actor
would select an
umelt composed of cognition actors one
level

below and how do we know that a a a a
particular level of cognition actors in
This growing hierarchy is complete it's
when all con actors one level below uh
belong to anelt

now it would not only grow but it would
also

shrink because whenever uh Cog actors
were recognized to be hopelessly
ineffective they would eventually be
removed and new cognition actors would
be would take their place and the hope
would be that uh as this cognition
hierarchy grows uh becomes more uh
competent we would have a robot that
starts acting uh intelligently and and
in survivable

ways so the Society of Mind would grow
shrink and evolve as needed to survive
now that's that's a key Point here um
this is an implementation of um a form

of mortal

Computing obviously the the robot is
physically the we Wheels uh the sensors
themselves uh are not mortal they're
they're like a a a a computer on Wheels
you turn it off you turn it on it's it's
still alive it's still it hasn't died by
from having been turned off however the
Society of Mind itself is Mortal it
grows it

evolves in such that the robot will be
competent at finding the food it needs
to keep the society of Mind alive so
what we have here is a um Immortal
Society of life of of mind trying to um
achieve a level sufficient level of
Competency to maintain
itself this is work in progress this is
a

um a system that I'm in the process of
building and it's composed of these
subsystems agency where uh the smart are
for growing shaping uh and evolving a
society of mind there's another
subsystem that it runs on the uh robot
itself which I call Body which provides
a um an interface to uh activate uh
Motors and to read
sensors there's also a a a version of
the body subsystem which will run in a
simulated world

and the subsystem world implements that
simulated environment it will also um
animate an a real
environment and then there's a subsystem
that I call Observer which provides me
or any uh other human being with a a
window into what's going on with the

within the Society of Mind of um the robot the agency subsystem runs on a powerful computer the the body subsystem the physical one runs on the brick pie which is powerful enough to run the uh significantly complex programs but not too computationally intensive ones and um world and observer and and body simulated run on a um a separate computer the Karma body subsystem as I said gives access to real and virtual sensors and effectors and here I have all the um the sensors I want to use and the motors

um removed from a a the robot just like connected directly to the uh Rick by and they are uh they can be uh activated they can be uh reached uh via an um arrest API which is essentially um they exposed a a a web interface uh to the agency subsystem the karma world subsystem uh here we see the simulated environment it's a grid

World

um the the gray tiles represent just the the surface and with a brown tiles representing obstacles so the the robot will bump into these obstacles and will perceive them with its uh simulated distance sensors the green patch is represents a Food Patch and the robot needs to find food in order to consume the food and replenish its U its resources if it runs out then it dies the uh we can see that some Tiles are are uh darker others lighter and

they this gradient this luminous
gradient corresponds to um um a a scent
gradient so the closer the robot is to
the food the lighter the tiles
are there will also be um poison patches
which will be red so detectable as a a
red surface on on the simulated floor
the robot must learn to avoid poison
food because that will harm it and if
it's armed uh if it's um
excessively harmed then it will also
die when a robot spends a enough time
over the food the food is consumed and
uh another Food Patch appears somewhere
else so the the environment is
dynamic and it changes with the uh with
the its interactions with the robot
that's an important
uh element here
now the um the karma agency subsystem is
the most complex and and the the key
subsystem and it implements a um
generative cognitive
architecture uh for the robot
implementing agency so we're going to
it's going to implement um cognition
actors as we've mentioned
already it will implement something I
call metacognition actors a very limited
uh understanding of the the the concept
of metacognition one metacognition
actor per level in the hierarchy and the
cognition actors are responsible for
adding removing uh conion actors as uh
needed as required then very very
importantly we'll have well-being actors
and these actors are represent the um
homeostatic values that

the society mind must keep in and within
a acceptable bounds in order to survive
we have fullness integrity and
engagement we'll talk in more details
about what they are
later and then very importantly uh
there's an app perception engine actor
uh last year at the Symposium I
presented the um my implementation of an
a perception engine and this is the um
very important part of
component of the Society of Mind it is
responsible for making sense of
observations in order to be able to make
predictions we'll go into more details
as
well now let's go and look at these
well-being actors they Implement uh
homeostatic uh States and I have three
one is fullness which represents how
much energy the uh the agent
as a whole the
robot has available to do things that
could be
Computing doing some comp uh expensive
computation or um just moving
about so there's a an energy budget that
gets
depleted and also gets replenished when
the robot finds food and spends enough
time in a Food
Patch there's also the Integrity
well-being actor and that represents uh
the health of the uh agent the if the
agent uh the
robot suffers too many collisions then
Integrity goes down and integrity is is
recovered over time think think the

think about lives in in a video game
that's pretty much the same concept and
finally a little bit bit more abstract
uh well-being actor is engagement and it
represents how to what degree the robot
is engaging
its environment we want to avoid the
dark room situation where the robot uh
stops doing anything uh stops moving
stop engaging its environment it h it it
has a it it it feels quote unquote an
imperative to to engage its
environment now the agent as a whole
meaning the Society of Mind of the agent
strives to maximize well-being for the
entire Society of mind now low
well-being levels will signal risk to
the survival of the Society of Mind so
that information will be available to
each individual cognition actor and to
the metacognition actors and will modify
uh constrained uh constraint what they
do key concept here is that well-being
imparts normativity to
beliefs beliefs acquired in the context
of a um low fullness or low Integrity
becomes a uh unpleasant undesirable
belief now this this this normativity
Associated to beliefs will focus
attention and motivate action as we'll
see later
on so well-being actors aggregate
metrics
uh received from the various cognition
actors and then aggregate these metrics
and broadcast uh wellness events back to
the cognition actors creating a give
essentially informing them of the um

ostatic State uh of the agent as a whole
now these matrix include the work that's
been done the food that's been
consumed these these two will impact the
fullness metrics

and values then uh poison consume
Collision count will impact the
Integrity overall Integrity of the agent
and then uh po predictions made
predictions received prediction errors
sent policies executed we'll see that in
more detail later all these are
indicative of the uh Society of Mind
engaging its

environment now metal cognition
actors their role is pretty
straightforward they're responsible for
adding cognition actors that are missing
and to remove underperforming cognition
actors each cognition actor is assigned
one level in the abstraction of
hierarchy and it looks at the conion
actors under its supervisions
supervision and and determines if all
the conion actors below them are part
of one or the other's

umelt so the notion of umelt here a
cognition

actor can only

observe or act within its umelt and what
is the umelt of a cognition actor it's
other cognition actors of a lower level
of

abstraction so at the the first level of
conion actors the prior level uh where
we have the um where they wrap sensors
and and

Motors um they don't they they don't

have a felt well they do it's the actual external environment of the robot they don't have an external anything above this

level uh the cognition actors have as part of their Umwelt other cognition actors and they constitute essentially the internal Umwelt of within the Society of

Mind so if a one layer of the um Society of Mind is complete it covers all all the cognition actors

below then it's time to create a cognition actor at the level above create a new layer of the Society of Mind and when you create a cognition actor you give it or the metacognition actor gives it its

Umwelt mostly

randomly with um the constraint that there's going to be overlap between Umwelt of various cognition actors and making sure that they have access uh to the motifs directly or indirectly now when

when well-being is low within the Society of mind when the well-being actors are reporting low wellbeing levels this might

accelerate the removal of cognition actors the metacognition actors being aware that we're not doing well now will say okay which cognition actors are underperforming our expectations and let's remove

them in in order to for them to be replaced with new C actors which hopefully will do

better and if uh well-being is low this will also slow down adding new cognition actors so this is an influence of well-being on the the actions of the Mion actors I want to say adding and removing um it's important to uh to to understand that adding is bottom up and removing is top down for example if I have a a meta if I have a cognition actor that has let's say two other conion actors in itsv I cannot remove a con actor in itsv because it depends depends on it but I can remove a cognition actor that is not in any other's umelt so that removal will be from the top to bottom and addition will be from the bottom to top what does a cognition actor do well a cognition actor observes its umelt of other cognition actors it generates beliefs from analyzing its observations and then it generates policies in order to modify or validate its beliefs and it does this this observe believe and act always within itself and one time frame at a time so each cognition actor goes through its more or less UDA Loop one time frame at a time now the the lower the lower a cognition actor is in the uh Society of Mind the more concrete it is the less abstract it is um the shorter this time frame will be it will cycle faster it's important also to realize that there there's no there's

um there's no synchronization between cognition actors each starts its time frame independently of other cognition actors and so that the the they may overlap but they don't coincide they don't overlap completely and the more abstract the cognition actor is the higher up it is in the hierarchy of cognition actors within the Society of Mind the longer its time frame will be which means that very abstract cognition actors will update their beliefs more slowly and will take actions

uh not as quickly as a a more concrete low-level cognition actor so we can see here the the cognition actor at the large one surrounded by its lowlevel cognition actors the what does it observe what does the con cognition actor observe in its unel well it observes it observes the belief beliefs of its of the cognition actors in itsv so as a more abstract cognition actor I observe the beliefs of the more concrete cognition actors in my umelt then based on these beliefs and as we'll see based on the normativity associated to the beliefs they good beliefs or they bad beliefs the cognition actor will uh um decide on what its goals are in this current time frame the goals would be modifying a particular belief or validating a particular belief then it will generate policies to attain this goal and then

emit this policy to its environment and say
okay do what I want and we'll see in
more details what do what I want
means So within each time frame
a cognition actor Will predict its Umwelt
after having made sense of it that's a
key uh aspect we'll we'll go into more
details it will update its beliefs by
analyzing past and present observations
so a cognition actor has a memory of
the current observations and this the
current time frame and observations made
in previous time frames and it will
update its beliefs but from analyzing
current observations and past
observations then it will select a goal
affecting a belief in it tends to impact
and construct a policy and recruit itself
to realize

it so if you look at the little U
redesigned UDA Loop here first there's
predict make predictions
about beliefs of other COG actors in
uh my Umwelt in order to predict I need
to have a causal model of my
observations that's what a perception
will do we'll see in more
details then the as a cognition actor
update my beliefs select goals act and
back to predicting in the next time
frame a perception is defined here as
the process of assimilating sensory
information into to a coherent unified
whole so let's look at uh
predicting what are the observations of
a cognition actor observations are
predictions it makes about the beliefs
of other cognition actors which are in

its

umelt

plus that plus prediction errors so it

makes predictions of itself of ition

actors in its umelt about their

beliefs and

the umelt cognition actors receive that

prediction look at it does it

corresponds to my current beliefs if yes

not saying anything I'm not contesting

the prediction if it doesn't match then

I emit a prediction error and the

cognition actor accumulates these

predictions that are uncontested and the

prediction errors and together they

constitute the Cog actor's current

observations

so we'll see that not all observations

receive the same attention that they're

not subject they're not equally subject

to prediction there's a prioritization

there's a notion of attention we'll see

that a bit

later attention attention prioritizes

which observations to update so the

cognition actor starts a time frame with

observ observations made in previous

time

frame and if there and it wants to make

to make those observations current to

the current time frame as many of them

possible but there's a limited amount of

time in which to um update these

observations so it the C actor will need

to prioritize which observations from

the previous time frame to update into

the more current observations so which

one will it try to update first well it

will look at
the
predictions that were
made by a higher level predic cognition
actors about its beliefs and knowing
which obs a support the beliefs will say
ah these is are the more these are the
observations that um more uh abstract
more important cognition actors are
making about me my state my beliefs so
I'm going to I'm going to try to make
sure that I have upto-date
observations that are of interest to my
parent my my parent cognition
actors so that's the first one I care
about the first observations I care
about the ones that are of interest to
my parent function actors actor sorry or
actors the second most important
observations in in terms of uh
updates are the observations that
support the strongest desirable and
undesirable beliefs remember that uh
beliefs are the result of analyzing
observations current and past if I have
very strong desirable or very strong
undesirable
beliefs then I will want to verify that
those OB observations are still current
so there will be the next uh
observations that I will try to uh
update then finally the the observations
that uh support weak or neutral beliefs
and then whatever remaining observations
they are so the conition actor will take
will prioritize the observations it it
it that existed from the previous time
frame and say okay I'm going to now make

predictions um that these observations are still valid or not valid and if those there are observation errors then then the the the observations are updated to the prediction error and if the observation is not contested then the observation is is um the one that is current this implements a predictive processing uh architecture because as you can imagine a high level cognition actor will make predictions about beliefs uh of the the the its children actors this will trigger predictions about the observation supporting disbeliefs one level down to the cognition actors in the layer below and all the way down so predictions flow down and prediction errors blow up through the entire hierarchy of uh the cognition actors in The Society of mind now in order to make predictions we need to have made sense of uh observations that's what app perception does so we need to make sense of observation in order to predict them now in order to to explain the concept of our perception in which in last year's presentation I went into quite a lot of details but just I want to give a an overall understanding here let's take the example of observing uh two lights light a and light B and the light can either be on or off and at different steps in time the lights may change their state from on to off off to on or

not so here we have a set of observations of light A and B over time and sometimes we may not be able to observe a light like uh light a in in the fourth state is a gray rectangle we don't know but we have a good enough set of observations and now the goal of the app Apperception engine is to find a causal theory that explains how the observations change over time why are we seeing some light a turn on after having been off or Etc in in order to formulate a causal Theory kind of a little program that when run recreates what we've observed the Apperception engine may need to imagine another light it might need to imagine some kind of connection between the lights so the Apperception engine will in in formulating a causal Theory will do some abduction some imagining of unobserved latent objects so we have a number of observations that are remembered from previous time frames we may uh we we don't accumulate all observations forever we forget the older ones and then we feed these observations into the app perception engine out of that comes a little program a causal theory that when when applied to the latest observation will produce the next

incoming expected
observations so that's a way of of of
making predictions of incoming uh
observations from current and past
observations from the current
observations given a causal theory so you
feed the latest observation you feed all
the observations into the app perception
engine it gives you a causal Theory you
feed the latest observation into the
causal Theory as input the output is
incoming predicted
observation so an Apperception
Apperception means finding a causal
explains by all the rules of the theory
and predicts
so a actor accumulates
observations wants to make sense of them
in order to predict incoming
observations and for other reasons as
well which we will see and it will what
it does is okay it it gives the app
perception engine actor its observations
over time and says please explain to me
what I'm observing give me a causal
Theory so I can predict future
observations given this causal theory if
if one is found successfully then the
contion actor can predict incoming
observations and what does it observe it
observes the beliefs of cognition actors
in itself more more concrete cognition
actors and then if the prediction is
correct if the causal theory is correct
and accurate we don't get a prediction
error if the causal theory is partially
correct some times correct then we get a
prediction

error so getting prediction errors
undermine confidence in the causal
theory that the Kung actor has received
from the aips and engine to explain what
it sees in its
umelt so it's quite possible that
there's a moment in time when the
observations when the causal theory is
is quite effective and it's accurate but
then something changes in in the state
of theel function actors such that the
causal theory is is is missing is is
very hit and miss and it may come a
point where the contion actor says I
don't like that causal Theory anymore
it's it's producing too many prediction
errors so it's going to ask the
perception engine actor give me a new
causal Theory based on the latest
observations that I've accumulated In My
Memory Okay so we looked at um how
cognition actors predict incoming
observations in order
to uh accumulate those new observations
to update those new observations based
on whether or not these uh predict the
predictions uh are contested by
prediction errors or not so now once we
have a new set of current uh
observations within this time frame that
we are operating in time to update
beliefs beliefs are if you want
um integrate are are the integration of
observations into a a a more
aggregated um
analytical uh assertion so cognition
actor updates its belief from analyzing
remembered observations and also uh

remembered actions that it that that
that it
took so and we'll see what kind of of
integration of observations and Analysis
of observations lead to what type of
beliefs but whatever type of belief it
is a belief will be desirable or
undesirable depending on how it
correlates with the agent's well-being
if a belief is being held mostly while
the a the agent as a whole is feeling
bad it has low resources or low
Integrity or or or limited engagement
then these these beliefs are going to be
tainted by the uh well-being
um levels of the agent at the time those
beliefs are
held now that's what makes a belief
desirable or undesirable what makes a
belief strongly held is if it is
supported by observations those
observations from which analysis the
belief is
derived if it is supported by
observations that are predicted by an
accurate causal Theory we have an
excellent causal Theory it's it's it
rarely misses
so um the the cognition actor is feeling
confident about its observations and it
makes the the whatever belief is is is
produced from analysis of these
successfully predicted observ ations
better supported more strongly
held now if a belief persists over many
time frames thanks to or in spite of
taken action taken to validate or
invalidate it if the belief

persists then it's even more strongly
held it's it's it's there's more there's
increased confidence that the belief is
uh represents uh what's really going on
in the umelt of the cognition

actor

now there are different kinds of beliefs
beliefs are are are generated are
produced from the analysis of uh current
and past

observations and okay so there's
abduction one type of beliefs is ISU
abduction and it's a direct consequence
of um having requested a causal Theory
and using a causal Theory remember in in
a causal Theory May um imagine a a
latent object or a latent relationship
between objects well that's that's a
form of belief I believe there's an
object that I don't see but must exist
in order for me for for a causal Theory
to make sense of actual observations I
made so an abduction belief would be
like I assume there's a hidden thing
next to the one I see

another kind of belief is just counting
I'm I'm I'm next to two things I have
made two observations that represent uh
say being next to something and so I
have two of them so I'm I can have a
belief that I'm next to two things
account belief now a trend belief is a
it's a belief that represents the the
the um discovery of a trend in a c
certain observations over time for
example example if um the con actor
observes distances to some obstacle and
those distances are getting

smaller

um over time then that's a trend my

distance to a thing is getting

smaller and then there's another kind of

belief is is some

observation ending

stopping by getting closer to a thing

stop

so that's an ending kind of belief and

then if the conion actor has uh taken

some action well there's the belief that

it tried to do something like I I tried

to stop my distance to a thing from

getting smaller so if the action was

trying to end the belief that the belief

in a trend I you can see now that

beliefs are are are composed they're

getting more and more abstract so I

tried to stop my distance to a thing

from getting smaller is a kind of

another kind of belief it's an attempt

kind of belief so we have all these

different kinds of

beliefs and when a Cogan actor

synthesizes the belief from analyzing

observations it assigns of belief a

globally unique predicate

name parent cognition actors the

cognition actors in which inelt this T

belongs um the parent cognition actors

observe predict the beliefs of cognition

actors in in in their

owns and reference them by name so if

the there's a belief in umel K actor

that some Trend

exists the um we're getting closer to an

obstacle well the this belief will be

exposed will be made available for

or observation to the parent as a named

uh

predicate why and how that belief uh

came about what the observations are by

the umel kin actor that led to that

belief it's it's it's opaque there's a

name and that's the name that's

available for observation there's an a

statement a predicate name predicate

that's available for observation by the

parent cognition

actor only the cognition actor that

holds the belief it named knows which of

its own observations support the belief

and how so if I say to you I

believe I I believe getting

closer that's something that can you can

observe as a parent con actor ah my

child con actor believes in getting

closer but the parent cognition actor

doesn't know why I believe we're getting

closer doesn't have access to the set of

observations that led to me holding the

belief that we are getting closer all

these distance observations that led to

me having holding the belief that

getting closer all that parent knows is

that I hold the belief called getting

closer and it's able to observe it

observe it you know ending or or

starting Etc but that's that's that's a

key um a key concept that beliefs are

are essentially

named um there are predicates their

names if you wish that uh hide the C the

reason why the belief is held to parent

contion actors that observe my beliefs

so there's information hiding going on

here

so a cognition actor observes the

beliefs of its children cognition actors

and

these these con these these beliefs that

are observed are the consequences of the

themselves observations about from of of

cons actors one level below so my belief

is is another if I'm a cognition actor

my belief is another cognition actor's

observation so it's really beliefs it's

it's beliefs about beliefs about beliefs

so it's beliefs all the way

down that's that's a maybe a little

tricky

concept if I'm at the top as a cognition

actor I observe beliefs of cognition s

under me these beliefs of connection

actors under me are supported by their

observations of beliefs of cognition

cognition actors under them so beliefs

about beliefs about

beliefs time to open a little

parenthesis here maybe take a

break the implementation that I'm

working on of

agency is a symbolic implementation so

it's its agency realized from symbolic

reasoning all the various agency actors

the the cognition actors the mid

cognition actors the well-being actors

the uh the perception engine actors all

run prologue programs prologue is a

programming language and those prologue

programs do symbolic

infome the cognition actor as we've seen

invents symbols invents gives names to

beliefs it discovers from analyzing

observation so it invents symbols and it
s synthesizes these beliefs as symbolic
statements getting closer to
XYZ the app perception engine as we've
seen very kind of
briefly its job is to discover causal
models and these caal models are
symbolic programs they're little logic
programs so agency is built on symbolic
reasoning in my
implementation closing the
parenthesis now let's talk about
precision and belief
strength the more accurate a causal
Theory the more precise it are are are
considered its predictions so we saw
earlier now precise predictions if
they're not contradicted by by
errors do strengthen the beliefs the
pred observations support so you have a
bunch of observations held by the
actor observations in the moment
remembered
observations these observations are the
result of predictions and or prediction
errors and we analyze these these these
observations coming from predictions to
generate beliefs I'm getting closer to
an obstacle from observations of
distances now if my observations of dist
es were predicted by a a really
top-notch causal Theory one that almost
never misses then I have confidence in
the beliefs that have derived from my
predictions I I feel confident that I am
indeed getting closer to an obstacle
given all the the the the distance
observations that I've accumulated

because I've accumulated them by making predictions using a very accurate empirically accurate causal Theory so precise predictions strengthens the belief that are derived from these from observations from predicted observations now the stronger beliefs are because I have may have strongly held beliefs and weakly held beliefs the strongly the more strongly held beliefs are the more attention they will receive When selecting when this the cogent actor needs to wants to select one and say I don't like this belief It's associated with poor well-being and I want to I want to change that belief I want I want to impact it I want to I want to get rid of that belief I want to make that belief not true or if it's a very pleasant belief oh I want to make sure that it is I want to validate it I want to keep that belief uh I want to keep holding that belief because it's a good one but I care whether this the belief is strongly held or weakly held because strongly held belief will get more attention will be the ones that'll be selected for validation or invalidation above above the the the weekly held beliefs so how do we impact beliefs that's the next step in the ud next two steps in the ud Loop of our cognition actor within each time frame it predicted based on its understanding of past observations it updated its belief from

uh from analyzing predict predictions
that were realized or or error s
predictions I those observations it
updated its beliefs and now it says it
needs to select a goal or may need to
select a goal that say I want to get rid
of this belief or I want to validate
this belief and how am I going to
accomplish this goal I'm going to select
the policy and then and then ask my
unal to act to realize this policy so
that's what we're looking at
now so a cognition actor acts via
policies it
formulates a cognition actor strives to
invalidate unpleasant beliefs and to
validate Pleasant beliefs but it will
prioritize strongly held beliefs whether
unpleasant or or
Pleasant a cognition actor will impact
the belief validate or invalidated it
holds by uh promoting or inhibiting the
observations that led to it now I have a
belief that we are getting closer to an
obstacle and somehow it's it's it's an
unpleasant belief maybe uh It's
associated with in the past with um
uh a decrease in in in U
Integrity leads to bumping into things
so I want to I want to put an end to
getting closer to an obstacle I want to
I don't want this belief that I'm
getting closer to an obstacle
to go away now I hold this belief
because I had observations of reducing
distances over time to some obstacle so
I want to change these observations
basically I want to I want to stop

getting close having um having uh
distances that get smaller I want to
stop this
so my observations of distances happen
to be beliefs
of PR actors in my umelt so what I'm
going to do I'm going to say okay I'm
going to select a number of of
observations that support the belief I
want to get rid of and I'm going to
formulate a policy as a list of goals to
my inelt CA I'm going to say I want you
to get rid of the fact that I have small
distances to an obstacle that's that's
that's your goal so I'm going to I'm
going to set a number of goals me the
actors who believes we're getting closer
to something and it's not doesn't feel
good I'm going to tell I'm going to
formulate a policy so that my umelt con
actors will try to um impact their own
beliefs that we're we have a small
distance to an obstacle for
example so that's that's a little bit
subtle here so I as a cognition actor I
want to modify a belief that belief is
based on observations these observations
are beliefs of umelt cognition actors so
I'm going to say to myel actors I want
you to to modify your own beliefs that
I've observed I want you
to change your beliefs beliefs that I'm
observing that lead me to believe that
I'm getting closer to an obstacle so I'm
setting up a bunch of goals for my umelt
ction actors so that if they realize if
they if they modify the these beliefs
then it will modify indirectly my own

belief that I'm trying to get rid
of okay so I'm basically telling my my
umelt these are the goals I want you to
realize so that I it I would indirectly
realize my own
goal Okay so this Pol a policy that I
put together are a set of goals for my
umel cognition actors to
realize
now a nelt conition
actor will then given the goal it
received formulates its own
policy because we're I'm
asking con actor in myel to
modify some some belief or other okay
says the con actor then I will formulate
my own policies to do that to to achieve
the goal that you that my parent
cognition actor sent to
me
so a cognition actor will receive
goals from parent con actors and these
goals received from IR up will have
higher priority that whatever goals the
cognition actor might want to formulate
on its own because the the umel function
actor has its beliefs has belief that it
wants to validate has belief it wants to
to to
invalidate and left on its own to its
own devices will do just that but if it
receives goals from a parent con actor
these take priority and it it will put
aside its own concerns to uh address the
concerns of the parent cognition
actor now a cog actor cannot execute only
one policy at a time which is why it
will prioritize the ones that it

receives from its
parent and when all is said and done the
cognition actor that initiated the
policy will determine will will verify
if this was successful or failure over
time if the belief they wanted to impact
was indeed impacted and a policy policy
that's known to work will be reuse
that's how con actors develop
habits okay

so where does the causal theory that the
con actor uh holds guide the formulation
of its policies because there was causal
causal causal Theory serves two roles
one is to predict incoming observations
and the other role is to formulate an
effective policy so let's say that
cognition actor is a belief that wants
to impact because uh

[Music]

the well-being me levels are are are low
Associated to that belief so he wants to
modify that that belief get rid of it
well it needs to formulate a candidate
policy which of the observations that I
made that I've analyzed into this belief
which which observations that support
this belief I want to
modify so which policy do I want to
formulate well well what it will do is
we'll say okay what do I need to do
given what my what I'm observing right
now so that I will observe something
differently that will
either remove the
belief or or reduce it so it will use
the causal Theory so what would happen
if I did X if I I I I I if I

asked my umelt to do
XYZ so the causal Theory will answer
that question we say
well plug that into the CLE Theory turn
the crank and this is what you would be
uh be observing next if you were doing
this if you if if you executed this
action this is what you would observe
next and then given what you would
observe next you can
calculate ah this these will be the
anticipated beliefs out of these
anticipated
observations might the policy work with
with the belief that I want to impact be
impacted the way I want to impact it if
so yeah let's use that policy otherwise
let's look for another one so the
cognition actor searches for an
effective policy given its understanding
of how things work in its umelt its
causal
Theory and and then applies the causal
the theory predicts what would happen if
it took this action if it if it
requested this goal to be realized looks
at the the the the resulting anticipated
beliefs and decides was this a a a good
effective policy and then and then it
would admit that policy and ask it umelt
please achieve these goals so that I
achieve my goal so and and that
percolates down because each cognition
actor being given a goal from the parent
Cog actor says ha how should I best
realize that goal what policy should I
formulate for my
own child my own

umelt in order

to realize the goal that was given to me
by my parent so it's policies all the
way

down and it bottoms out when at the
level where uh effects are produced
moving Motors turning moving forward
moving backward

whatever okay so that's a lot of uh a
lot of content describing this cognitive
architecture this generative cognitive
architecture what I hope will happen is
through this growing shrinking evolving
um um refining of this Society of Mind
composed of of cognition actors
um with uh in the context of of of of
homeostatic measures provided by
well-being

actors I'm I'm hoping that with policies
going down beliefs percolating up
predictions going down predictions going
up all this this this this changes and
evolution of the Society of Mind all
these all these activities from all of
this would

emerge agency and I I'm looking at that
as as a form of operational closure with
uh constraints flowing down constraints
flowing up and eventually hopefully the
um the Society of mine will explore a
space of

possible organizations and and and and
and and and space of of possible uh ways
of defining cognition actors relating
cognition actors together such that it
leads to behaviors of the robot
that preserves the life of the Society
of Mind that Society of Mind never

reaches a point where it runs out of
fuel where it is damaged Beyond
repair that's that's the overall
goal I think of the Society of Mind as
kinda autopoietic with all these
um elements of the Society of Mind
leading to other elements for example a
causal Theory will lead to predictions
predictions will lead to prediction
errors prediction errors will lead to
observations uh observation will be fed
into a perceptions which will lead to an
updated causal Theory and and
observations lead to to beliefs um
beliefs lead to
policy um a CLE Theory will determine
which policy is more likely to work
policy leads to effect effects lead to
New Sensations Sensations lead to new
observations and so forth and so on so
this I look at this as a a kind of a
um a
a set of of of processes that support
each other and in in something of a of a
closed set of complex Loops leading to a
Society of Mind that survives over
time now there's a number of pH
philosophical stenes that are taken by
this project and by philosophical stance
I mean uh practic pragmatically
Justified perspectives a way of seeing
in the
world first I look at active inference
as a normative framework I don't think
that's controversial at all I mean
that's that's what it's defined to be
and uh to persist an agent must behave
as if it acts to minimize surprisal now

clearly I'm not
implementing the uh Society of mine
using the mathematics of the free energy
principle but I'm implementing it
hopefully such
that doing an active inference analysis
of the robots observable behaviors in
states will demonstrate that the uh
Society of Mind the agent as a whole is
acting such as to reduce
surprisal so I'm
implementing an agent and also its
environment as a system of generative
processes the active inference analysis
will be done in terms of generative
models that are inferred uh from
observing the behaviors of the robot and
uh observing the states that are visible
to an outside
Observer other uh philosophical stances
I'm just going to go through them very
quickly each one is worth an entire
presentation I I I take the sense that
cognition arises from an EMB Ed agent
engaging its environment that sense
making is the discovery of unified
causal theories that meaning is grounded
in the agent's drive to survive no
mortality no meaning that intelligent
emerges from interactions between a
collective of
parts that's the Society of mind and
that the parts constrain the hole and
the hole constraints the part so
constraint closure
I'm implementing I'm in the process of
implementing this this this this system
that will hopefully demonstrate

artificial

agency and here's the current progress

there are three phases in this project

phase one I developed an cognitive

architecture that's what I presented

today uh implemented a one version of

the the app perception

engine and that's what I presented last

year I implemented the subsystem to uh

realize a to connect to sensors and

actuators that's the Cara body and I

implemented a simulated grid World in

which a simulated um agent would would

operate and and and interact that's to

uh speed up uh implementation because it

takes a long time to uh if I were to do

this only in the real world the um the

the edit run debug Loop would be

extremely

long and so I'm in the process right now

of implementing uh the agency uh

capabilities in phase two um which would

be about uh six months to a year from

now I will have a a redesigned robot

body I will um

uh be testing in a real life

environment and I will be collecting a

lot of data through a a The Observer

subsystem and in the third and final

phase the uh data gathered from runs

both in real life and in the virtual

world will be subjected to active

inference analysis to validate that my

agent does survive because it minimizes

Sur

risal that's it um if you are interested

you can contact me at uh this email

address JF do c Lou T ractive inf.

Institute you can also uh reach me on uh
the Sim Cog robotic channel of the
institutes

Discord and if you're interested in the
source code it is on GitHub so I'd like
to uh thank uh the active INF Institute
for uh support and encouragement